

Expected Direct Search Charged Lepton Flavor Violation Sensitivity in μ^+ Decays at Rest

The Mu2e Collaboration

We examine the sensitivity of the Mu2e experiment to a probe of Charged Lepton Flavor Violation in two-body $\mu^+ \rightarrow e^+ X$ decay. Although Mu2e is primarily a search for $\mu^- N \rightarrow e^- N$ the experiment can be easily modified to accept μ^+ instead of μ^- . The proposed μ^+ run measures the e^+ momentum spectrum from 3×10^{13} μ^+ decays at rest, below the Michel edge $E_e \leq 53$ MeV. Mu2e can establish 90% C.L. branching ratio limits for $\mu^+ \rightarrow e^+ X$, where X is a new light particle, over the mass range $20 \leq m_X \leq 50$ MeV. The most sensitive limit is at $m_X \approx 20$ MeV with $\text{BR} \leq 1.3 \times 10^{-7}$ (90% CL), improving the current experimental limit by two orders of magnitude.

I. INTRODUCTION

Phenomena not accounted for by the Standard Model motivate searches for physics beyond the Standard Model (BSM). Charged lepton flavor violation (CLFV) provides a particularly sensitive probe of such physics [1–6]. The most intensively studied CLFV channels involve transitions between muons and electrons: $\mu^+ \rightarrow e^+ \gamma$ [17, 18], $\mu^+ \rightarrow e^+ e^- e^+$ [19–22], and coherent $\mu^- N \rightarrow e^- N$ conversion [23, 24].

The Mu2e experiment [25] at Fermi National Accelerator Laboratory (Fermilab) is a search for coherent $\mu^- N \rightarrow e^- N$ conversion. The process occurs when a stopped negative muon is captured into an atomic orbit of the target nucleus and converts to an electron. Mu2e targets an unprecedented single-event sensitivity of $\mathcal{O}(10^{-17})$ [26]. The analysis focuses on the conversion-electron (CE) signal at the kinematic endpoint, $E_{\text{CE}} \simeq 105$ MeV, where Standard-Model (SM) backgrounds are strongly suppressed. The dominant background arises from muon decay in orbit (DIO); near the endpoint its spectrum falls as $(E_{\text{CE}} - E)^5$, broadened by detector resolution, within the signal window [56].

In the absence of a CLFV signal, or with limited signal statistics, the DIO background lacks distinctive spectral features that can calibrate the momentum scale and shape. Consequently, Mu2e considers dedicated calibration runs, including measurements of stopped $\mu^+ \rightarrow e^+ \nu_e \bar{\nu}_\mu$ decays, stopped $\pi^+ \rightarrow e^+ \bar{\nu}_e$ decays, and dedicated cosmic-ray datasets. The reversal only requires rotating a collimator that selects one of μ^- or μ^+ [57].

A calibration run using μ^+ decay at rest would measure the well-known Michel spectrum [27–32]. Operating at half the nominal magnetic field, this run would yield positrons from the sharp Michel endpoint at $E \simeq 53$ MeV with helical trajectory parameters similar to those of the CLFV $\mu^- \rightarrow e^-$ conversion electron (CE) at full field.

In parallel, a data set of $\mathcal{O}(10^{13})$ stopped μ^+ would enable a direct CLFV search for the two-body decay at rest $\mu^+ \rightarrow e^+ X$, where X denotes a new light particle with mass $0 \leq m_X \leq 50$ MeV. The two-body μ^+ decay produces a monoenergetic positron, with the positron energy fixed by the muon mass and m_X . Such searches test a

broad class of CLFV models [33–50]. This paper presents the Mu2e sensitivity for $m_X \geq 20$ MeV; lower mass particles are progressively more difficult to disentangle from the Michel peak.

In addition, the μ^+ decay-at-rest data set provides sufficient statistics to reduce key systematic uncertainties, including the magnetic-field strength and the momentum response function. These μ^+ -based systematic improvements directly enhance the overall sensitivity of the BSM $\mu^- \rightarrow e^- N$ search.

This run requires operation of the detector solenoid (DS) magnetic field at $\approx 50\%$ of its nominal value (52.8 MeV/104.5 MeV) to ensure high acceptance for Michel positrons entering the tracker (see Fig. 1.) However, lowering the magnetic field comes with a complication: the Mu2e detector is annular, with an inner radius chosen so that there is minimal acceptance for positrons at the Michel edge following their helical orbits in the 1 T solenoidal field. Lowering the magnetic field increases the radius of the helices and the acceptance of those positrons. This in turn results in excessive detector occupancy, requiring a substantial reduction of the proton beam intensity. This reduction is required not only to limit the occupancy (dead-time considerations) but also to limit the number of reconstructed tracks per event and to remain within the data-throughput constraints of the DAQ system. Detailed simulations indicate events with up to ≈ 10 tracks can be processed with only a 10% loss in the reconstruction efficiency. Accordingly, in the following discussion, we present results for two limiting scenarios: low-reach ($\times 1000$ beam intensity reduction, 2–3 tracks per pulse, and high reach ($\times 100$ beam intensity reduction, 20–30 tracks per pulse. Assuming two weeks of continuous μ^+ running (336 wall-clock hours with a macroscopic $\sim 37\%$ duty factor) corresponds to 3×10^{12} and 3×10^{13} μ^+ stops respectively.

II. MU2E DETECTOR

The primary physics objective of Mu2e is the search for coherent $\mu^- N \rightarrow e^- N$ conversion on an aluminum target. The Mu2e apparatus comprises three graded super-

conducting solenoids, a charged-particle tracking system, and a calorimeter, as detailed in Ref. [26].

Mu2e will produce muons using an $E_{\text{kin}} = 8$ GeV proton beam extracted from the Fermilab Delivery Ring at a $1.695 \mu\text{s}$ repetition rate. The beam strikes a (tungsten for these simulations) production target located at the center of the superconducting production solenoid (PS), generating pions. The graded magnetic field of the PS reflects forward-produced charged particles; these, together with backward-produced particles, are transported into the S-shaped superconducting transport solenoid (TS).

Mu2e's muons will be produced primarily through pion decays occurring in both the PS and TS. The curved magnetic field of the TS induces vertical charge-dependent drifts, separating positively and negatively charged particles. At the location of maximal separation, a vertically offset, rotatable collimator selects either negative or positive particles. This collimator is the only piece of the apparatus that needs to be changed to switch from directing μ^+ instead of μ^- to the detectors.

The superconducting detector solenoid (DS) contains an upstream graded-field region that houses the muon stopping target. The downstream region provides a uniform 1 T magnetic field in which the straw-tube tracker and undoped CsI crystal electromagnetic calorimeter will be installed. Approximately one-third of the muons entering the DS stop in the target. After stopping, negative muons undergo atomic capture and rapidly cascade to the muonic $1S$ orbit in aluminum. These bound muons may undergo coherent $\mu^- N \rightarrow e^- N$ conversion, producing a monoenergetic conversion electron (CE) with energy $E \simeq 105$ MeV.

These same stopped muons bound to aluminum atoms can undergo decay in orbit (DIO), producing electrons with a momentum spectrum that extends to 105 MeV but falls rapidly near the endpoint. The Mu2e detector is annular; in full-field operation, electrons (or positrons) at the Michel peak have a radius of curvature less than the radius of the annulus. To suppress the sharply increasing low-radius occupancy and mitigate radiation damage to detector components, Mu2e leaves the inner detector radius uninstrumented. Consequently, the tracker reconstructs only electrons emitted from the stopping target with momenta $p_e \gtrsim 70$ MeV.

A key feature of nominal Mu2e operation is the pulsed proton beam with a $1.7 \mu\text{s}$ spacing, matched to the 864 ns lifetime of bound μ^- in aluminum[51]. The “good data” window opens at ≈ 500 ns after the arrival of a proton pulse; before this time beam flash (largely produced by π^0 's born in the proton target; these π^0 's decay to photons that subsequently convert to e^+e^- pairs that are then transported through the solenoid system) and muon-capture byproducts generate approximately 2000 hits in the tracker. Michel electrons do not contribute, as at full magnetic field they traverse the uninstrumented central region of the helical tracker. The straw-

tube tracker measures the conversion-electron momentum and, together with the electromagnetic calorimeter, provides particle identification that suppresses backgrounds from through-going muons to negligible levels.

Fluctuations in energy loss within the stopping target and the proton absorber ring upstream of the tracker widen the CE peak from the tracker resolution of ≈ 180 keV (FWHM) to ≈ 950 keV at $p=105$ MeV.

III. VIEWING THE MICHEL μ^+ DECAY SPECTRUM

A dedicated μ^+ run with suitable statistics enables searches for new physics in channels not probed by the coherent muon-to-electron search at Mu2e. As explained in Sec. I the 50% lower field exposes the detector to the Michel spectrum, requiring:

- (i) the proton beam intensity to be reduced by a factor of $\mathcal{O}(300)$,
- (ii) an acceptable number of tracks per pulse (7–10) need to be reconstructed, and
- (iii) the streaming data throughput remains within the capability of the data-acquisition system.

This configuration provides both a precise detector calibration using the e^+ Michel spectrum and sensitivity to a BSM search for the decay at rest $\mu^+ \rightarrow e^+ X$ in the mass range $0 < m_X < 50$ MeV. The data sample is based on a stopped μ^+ sample of 3×10^{12} and 3×10^{13} muons.

A simulated Michel spectrum and its differentiated form (from which we extract the resolution) are shown in Fig. 1, where the Standard Model prediction is convoluted with the expected detector response. The narrow peak in the detector response shown in Fig. 1, with an energy resolution $\Delta E \approx 1$ MeV, is well suited for searches for positrons produced in two-body final states, $\mu^+ \rightarrow e^+ X$, and provides strong background suppression in the signal region. The peak at 52.3 MeV in Fig. 1 (lower) is shifted downward and widened from the free-muon endpoint at 52.8 MeV because of the stochastic energy loss in and various paths through the stopping target. It is representative of the expected signal shape for an $m_X = 0$ muon decay. In such a search, Michel positrons dominate all other background sources in the tracker by approximately two orders of magnitude; all other background contributions are therefore small and are neglected in this study.

A dedicated μ^+ run collecting $\mathcal{O}(10^{13})$ μ^+ stops enables both a precise detector calibration based on the observed e^+ Michel spectrum and a high-sensitivity search for a BSM particle X in the decay-at-rest $\mu^+ \rightarrow eX$ within the mass range $0 < M_X < 50$ MeV. The choice of this mass range is explained below in Sec. IV.

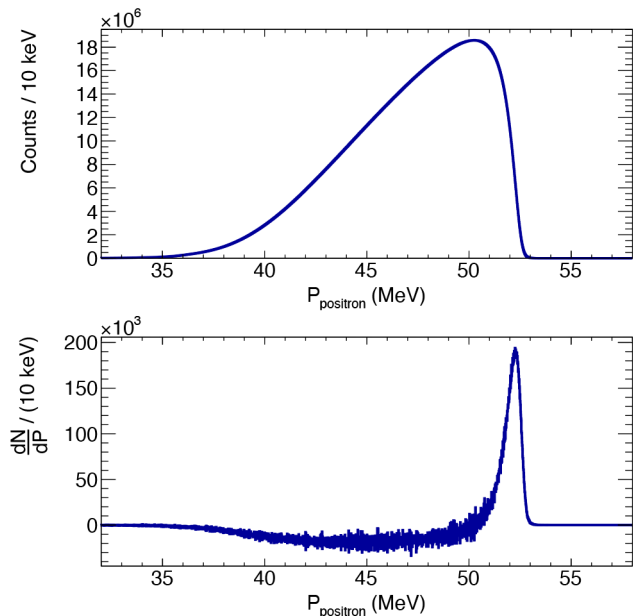


FIG. 1. (Upper) Simulated Michel positron spectrum after track reconstruction and selection, statistically enhanced using a fit function. The spectrum corresponds to 2×10^{11} stopped μ^+ decays. (Lower) The differentiated spectrum, dN/dp , showing the peak at 52.3 MeV corresponding to the reconstructed Michel-edge momentum. The full width at half maximum (FWHM), $\sigma_{\text{FWHM}} \approx 900$ keV, provides a measure of the expected resolution for a zero-width anomalous signal.

IV. $\mu^+ \rightarrow e^+ X$ SEARCH WITH $m_X \geq 20$ MeV

Using the μ^+ decay data set to search for the direct CLFV two-body decay at rest $\mu^+ \rightarrow e^+ X$ corresponds to a search for the emission of a monoenergetic positron, with kinematics constrained by $m_X \leq m_\mu - m_e = 105$ MeV. Setting a limit on $\text{BR}(\mu^+ \rightarrow e^+ X)$ therefore reduces to a search for a narrow spectral peak, similar in shape to that shown in Fig. 1, superimposed on the Michel spectrum.

For a given mass m_X , the uncertainty $\delta\mu_0$ on the number of events μ_0 in the signal window consists of a statistical component, $\delta\mu_{\text{stat}}$ and a systematic component, $\delta\mu_{\text{syst}}$, added in quadrature.

The search employs two distinct methodologies to cover the full accessible momentum range: one for $m_X \geq 20$ MeV, where statistical uncertainties dominate, and another for $m_X \leq 20$ MeV, where systematic uncertainties dominate. The boundary at $m_X = 20$ MeV corresponds to a positron momentum of $p_e = 50.6$ MeV. For $m_X < 20$ MeV, the signal positron momentum exceeds 50.6 MeV, placing the search for a distinctive spectral peak in close proximity to the Michel edge, where systematic uncertainties dominate. Consequently, sensitivity in this region is significantly reduced and requires more dedicated studies of reconstruction effects.

In this work, the search region has therefore been

restricted to the region where statistical uncertainties are the dominant contributions, allowing the use of a sideband-based search strategy.

For the decay at rest $\mu^+ \rightarrow e^+ X$ with positron momentum p_{e^+} in the reconstructed range $p_{\text{reco}} \leq 50.6$ MeV, well separated from the Michel edge, a sideband search strategy is employed. The uncertainty in the signal window is constrained by fitting the momentum spectrum outside the signal window. The sideband strategy relies on the fact that spectral-shape variations induced by individual systematic sources are smooth on the scale of the $\mu^+ \rightarrow e^+ X$ signal width, allowing systematic uncertainties to be neglected and rendering the search sensitivity statistics limited. In this work, the “observed” positron spectrum S_p^{obs} is generated using Monte Carlo simulation, employing the full Mu2e detector simulation and reconstruction algorithms. For a given m_X , the spectrum is fitted (using *MINUIT*[55]) for each signal amplitude a using

$$S_p^{\text{obs}} = (1 - a)f_{\text{SM}} + af_{m_X}^{\text{signal}}. \quad (1)$$

f_{SM} is the fit spectral shape of the SM background, fixed to the expected shape of S_p^{obs} . $f_{m_X}^{\text{signal}}$ is the spectral function of the e^+ signal of mass m_X . Both spectral shapes are normalized to $\int S_p^{\text{obs}} dP = 1$. The momentum line width of the positron signal is assumed to be negligible compared to the ≈ 1 MeV FWHM detector momentum resolution. The zero line-width assumption is valid for any X particle with a lifetime $\tau \geq 10^{-21}$ seconds.

Thus the only fit parameter is the signal amplitude, a , in the asymmetric 1.5 MeV momentum search window, $(p_{\text{peak}} - 1 \text{ MeV}, p_{\text{peak}} + 0.5 \text{ MeV})$. The choice of signal window is somewhat arbitrary and based on the asymmetric signal shape after dE/dx losses; see [26], Fig. 11. The precise values of the size and asymmetry are momentum dependent and we have chosen a representative value.

For each m_X one fit is performed by fixing the S_p^{obs} histogram at its expectation and passing the data statistical uncertainty on the histogram bins to the fitter. The signal amplitude fit parameter a has an expectation value of zero, while its fitting uncertainty δa becomes the measure of uncertainty $\delta\mu_0$ in the null signal hypothesis.

The 90% confidence-level branching-ratio sensitivity for $\mu^+ \rightarrow e^+ X$ with $20 \leq m_X \leq 50$ MeV is then estimated. The result is shown in Fig. 2, where the curves correspond to data taking with 3×10^{12} stopped muons (upper curve) and 3×10^{13} stopped muons (lower curve).

For the expected data set, the sensitivity lies between these two extremes. In the case that 3×10^{13} stopped μ^+ are achieved, the 90% confidence-level branching-ratio sensitivity for massive X particles in the mass range $20 \leq m_X \leq 50$ MeV reaches approximately 1×10^{-7} . This limit is approximately two orders of magnitude more sensitive than the current world-best result from

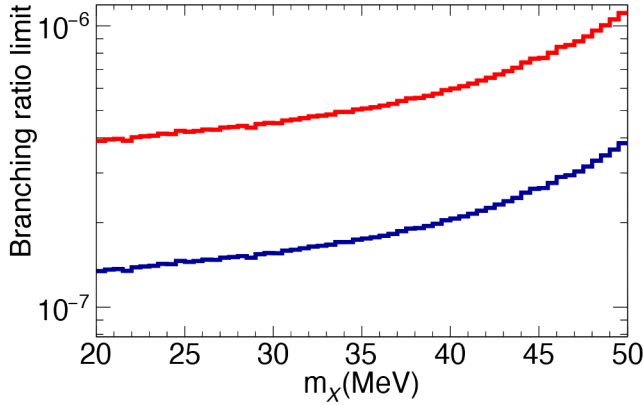


FIG. 2. 90% confidence-level branching-ratio limit for $\mu^+ \rightarrow e^+X$ as a function of m_X , assuming 3×10^{12} stopped μ^+ at a proton-beam intensity reduced by a factor of 10^3 relative to nominal (upper curve) and 3×10^{13} stopped μ^+ at a proton-beam intensity reduced by a factor of 10^2 (lower curve). In both cases, statistical uncertainties dominate over the momentum search window.

the TWIST experiment, which reports an average 90% confidence-level branching-ratio limit of 1×10^{-5} [54].

V. CONCLUSIONS

The Mu2e indirect CLFV search experiment [25, 53] is optimized to observe the coherent conversion reaction $\mu^-N \rightarrow e^-N$. The decay-in-orbit (DIO) background in the signal window lacks spectral features that can be used to verify the shape and scale of the measured momentum spectrum. A separate calibration run using $\mu^+ \rightarrow e^+\nu_e\bar{\nu}_\mu$ decays at rest is therefore used to determine the spectrometer settings. Such a run allows approximately 3×10^{13} μ^+ decays at rest to be collected over a period of $\mathcal{O}(2)$ weeks of data taking.

In parallel, the μ^+ calibration data set enables a direct CLFV search for the two-body decay $\mu^+ \rightarrow e^+X$, where X denotes a new light particle. Over the mass range $20 \leq m_X \leq 50$ MeV, the expected branching-ratio limit reaches $\text{BR} \simeq 1.3 \times 10^{-7}$ at 90% CL at $m_X \approx 20$ MeV, improving the current world-best constraint by approximately two orders of magnitude.

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